

RIDWAN AMURE

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EDUCATION

University of Arkansas - Little Rock

PhD, Computer Science

Expected May 2027

MS, Computer Science

May 2025

EXPERIENCE

Meta

June 2026 – Aug 2026

Software Engineering Intern (Ads Ranking (Early Stage Ranking))

- Applied product quantization to the Early Stage Ranking (ESR) model, enabling GPU-based serving at billions-of-candidates scale with improved inference latency.
- Developed a GRPO-based auxiliary loss for ESR, adapting the group relative policy optimization objective from LLM post-training to improve ranking model optimization without supervised labels.
- Designed hard target-aware attention via ID matching, delivering measurable gains in Normalized Entropy (NE) and recall across the ESR pipeline.
- Introduced joint retrieval with in-batch softmax loss for ESR, improving training signal efficiency for large-scale candidate retrieval.
- Resolved model-lowering and inference compilation blockers in GPU-deployed ranking models, unblocking production machine serving.
- Improved overall ESR architecture through systematic optimization of the candidate interaction layer across the retrieval stack.

Meta

May 2025 – Aug 2025

Software Engineering Intern (Machine Learning)

- Built an end-to-end ML data pipeline that reduced training data latency from 14 days to 1 day, enabling near real-time model iteration.
- Improved large-scale ranking model quality through feature engineering, ablation studies, and integration of high-impact signals.
- Designed a transformer-based interaction architecture for retrieval and ranking, optimizing inference-time performance at scale.
- Developed model-based retrieval systems to improve candidate generation quality for large-scale ranking pipelines.

University of Arkansas - Little Rock

Aug 2023 – Present

Researcher

- Built COSMOS, a large-scale AI pipeline for cross-platform social media analysis, processing millions of posts across datasets with 100K+ users
- Developed knowledge graph and embedding-based models for narrative tracking and entity representation across platforms
- Designed ML systems for modeling information diffusion and behavioral dynamics using graph-based and statistical approaches
- Implemented scalable data pipelines for preprocessing, feature extraction, and modeling on high-dimensional datasets
- Applied network science and machine learning to extract actionable insights from complex relational data

IQuartic (Newwave Company)

Jan 2023 – Aug 2023

Senior Machine Learning Engineer

- Developed end-to-end ML systems for ICD-to-HCC risk adjustment, increasing financial gains by over 30% through improved coding accuracy
- Built Bayesian inference models to identify missed or suboptimal medical coding in electronic health records
- Designed NLP and RAG pipelines for structured extraction and analysis of clinical text data
- Deployed production ML models with full lifecycle management, including training, validation, and monitoring

ParallelScore

Sep 2021 – Apr 2023

Data Science Engineer

- Developed transformer-based NLP models for medical coding tasks, including PHI, ICD extraction, and visit classification with improved PHI detection performance from < 50% to 95% in QA, and 89% accuracy in production deployment
- Built scalable ML pipelines for processing and extracting structured insights from large-scale unstructured clinical text
- Designed and deployed inference systems using AWS, Docker, and Kubernetes for real-time model serving
- Conducted model evaluation, error analysis, and performance optimization for production-grade AI systems

Babban Gona

Sep 2020 – Sep 2021

Machine Learning Engineer

- Built computer vision models deployed to over 200,000 users for agricultural diagnostics
- Developed low-latency inference systems for real-time detection of plant conditions
- Maintained and optimized production ML systems in resource-constrained environments

SKILLS

Python, PyTorch, TensorFlow, NLP, Transformers, Knowledge Graphs, AWS, Docker, RAG, Agents, Agentic AI